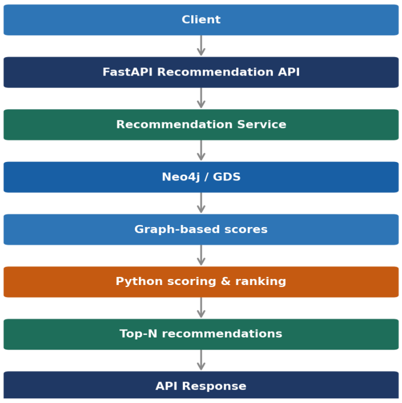
Recommendation Module Architecture

Graph Data Science — Student Social Network · GDS-21

1. Objective

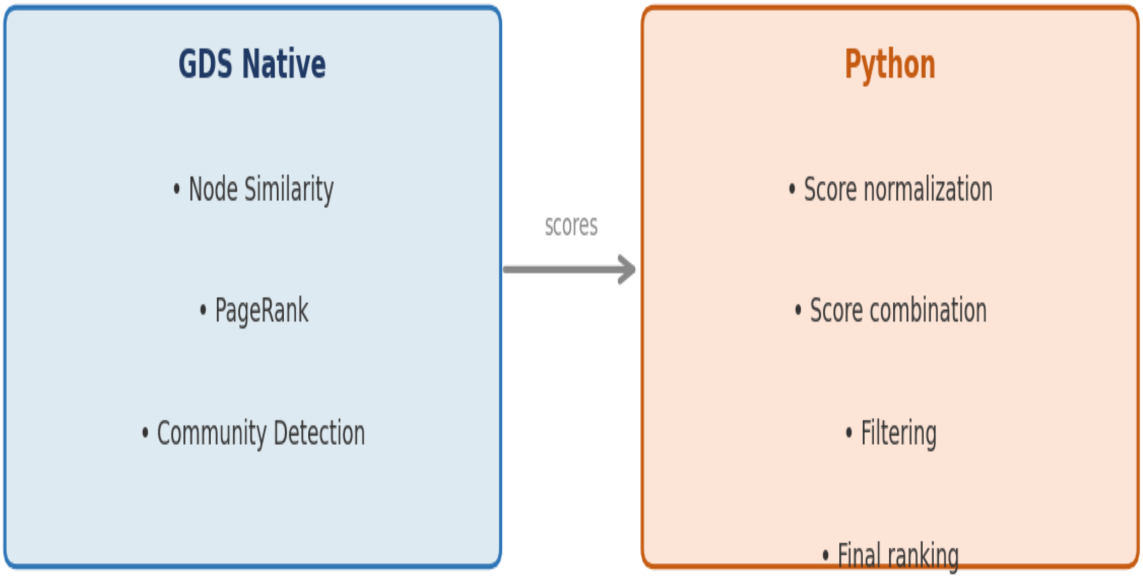
The recommendation module provides personalized recommendations for students based on graph relationships and graph-based algorithms. It combines the native computation power of Neo4j Graph Data Science (GDS) with Python-side scoring and ranking to deliver a Top-N list of relevant students, clubs, or events through a REST API endpoint.

2. Data Flow



3. Computation Strategy

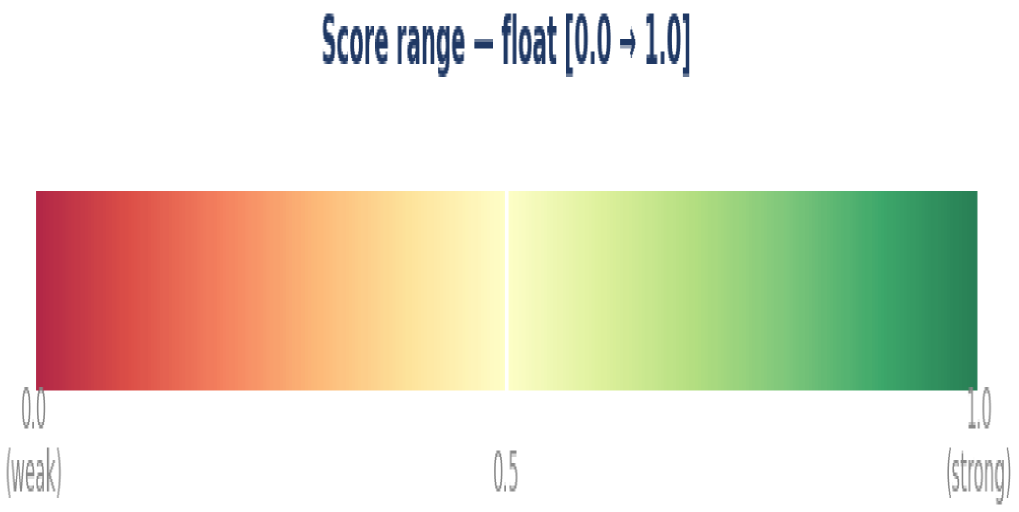
The module splits the computation workload between two layers. Neo4j GDS handles graph-native operations that benefit from the native graph engine. Python handles post-processing steps that require flexibility and composability.



Layer	Responsibilities
GDS Native	Node Similarity · PageRank · Community Detection
Python	Score normalization · Score combination · Filtering · Final ranking

4. Score Format

Property	Value
Type	float
Range	0.0 — 1.0
Meaning	Higher score = stronger recommendation



5. API Endpoint

The recommendation service is exposed through a single REST endpoint. The client sends a student ID and receives an ordered list of recommended students with their computed scores.

Endpoint

```
GET /api/v1/recommendations/students/{studentId}
```

Response format

```
{
  "studentId": 13,
  "recommendations": [
    {
      "studentId": 27,
      "score": 0.91
    },
    {
      "studentId": 44,
      "score": 0.78
    }
  ]
}
```

Response fields

Field	Type	Description

studentId	integer	ID of the requesting student
recommendations	array	Ordered list of recommended students
recommendations[].studentId	integer	ID of the recommended student
recommendations[].score	float	Recommendation score between 0.0 and 1.0